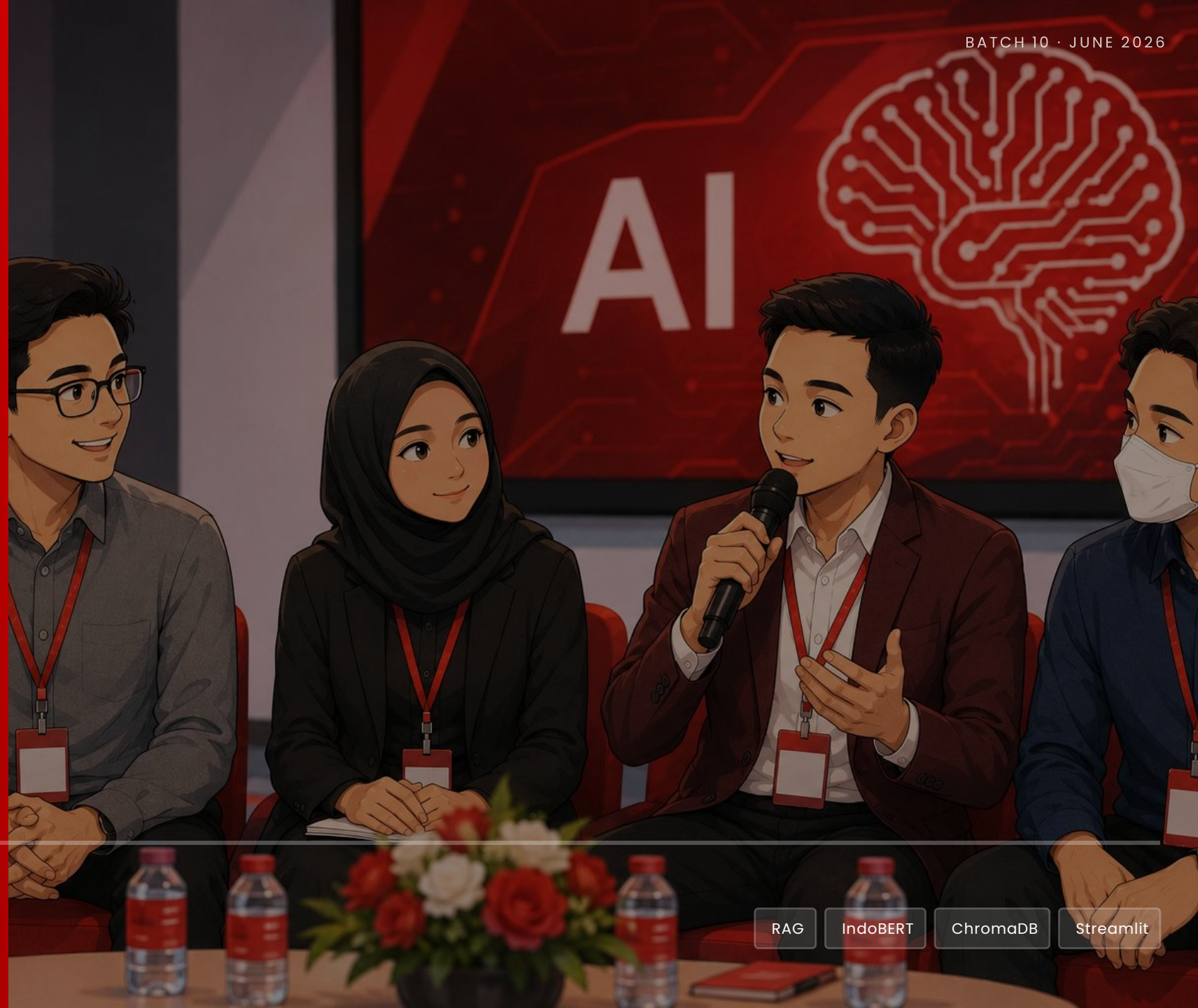


— RETRIEVAL-AUGMENTED GENERATION

News Intelligence *Assistant*

A grounded question-answering system over Indonesian news — combining semantic retrieval, category routing, and sentiment analysis to deliver source-cited, anti-hallucination answers.

NLP - Group C



RAG

IndoBERT

ChromaDB

Streamlit

Too much news, too few *trustworthy answers*

Indonesian online media publishes a relentless stream of articles every day. Finding a reliable answer — and knowing where it came from — is slow, manual, and easy to get wrong.



Information overload

Thousands of articles span finance, sport, tech, health, and general news — far more than any reader or analyst can keep up with.



Keyword search isn't enough

Traditional search returns a list of links, not a direct answer, and misses semantically related coverage worded differently.



Ungrounded AI hallucinates

Generic chatbots invent plausible-but-false answers with no sources — a real risk when decisions depend on accurate information.

THE OPPORTUNITY

A system that reads the news **for you** — answering questions in natural language, **grounded in real articles**, with citations you can verify.

Add category routing and sentiment, and the same engine becomes a lens for monitoring issues and public mood by topic.

A grounded assistant for *Indonesian news*

WHAT IT IS

A News Intelligence Assistant — a question-answering system over real Indonesian news articles, built on **Retrieval-Augmented Generation (RAG)**. It fuses semantic search, category classification, and sentiment analysis to answer with traceable, cited sources.

APPROACH

- ✓ Scrape 2,000 Detik articles across 5 channels
- ✓ Fine-tune IndoBERT for category & sentiment (SmSA)
- ✓ Chunk + embed (multilingual MiniLM) into ChromaDB
- ✓ RAG: retrieve + metadata filter → LLM with grounding

OUTCOME

- ✓ Full RAG news Q&A as a Streamlit application
- ✓ Category classifier (~96% acc) & sentiment model
- ✓ Vector database of 4,448 searchable chunks
- ✓ Grounded, source-cited answers with filters

WHY IT MATTERS

- ✓ Accelerate access across many articles at once
- ✓ Trustworthy, source-based answers reduce misinformation
- ✓ Monitor issues & public sentiment per category
- ✓ Foundation for an AI media-intelligence product

Why it matters — and the *value it creates*

Beyond a working prototype, the assistant delivers concrete operational value: faster knowledge access, a guardrail against misinformation, and a foundation for an AI media-intelligence product.



Hours → seconds

FASTER ACCESS

Collapses manual reading of many articles into one source-cited answer, on demand.



100% grounded

TRUST & ACCURACY

Every answer is drawn from real articles with citations; the system says "not found" instead of fabricating.



5 categories

MONITORING

Track issues and public sentiment per topic to inform comms, product, and risk decisions.



Together these form the foundation for an **AI media-intelligence product** — extensible to more sources, live updates, and enterprise monitoring.

Group C — four contributors

A cross-functional team spanning data collection, model fine-tuning, retrieval engineering, and application delivery for the end-to-end RAG system.



Agil Setiawan

GROUP C · NLP



Khaerani Arista Dewi

GROUP C · NLP



Surya Dharma Putra

GROUP C · NLP

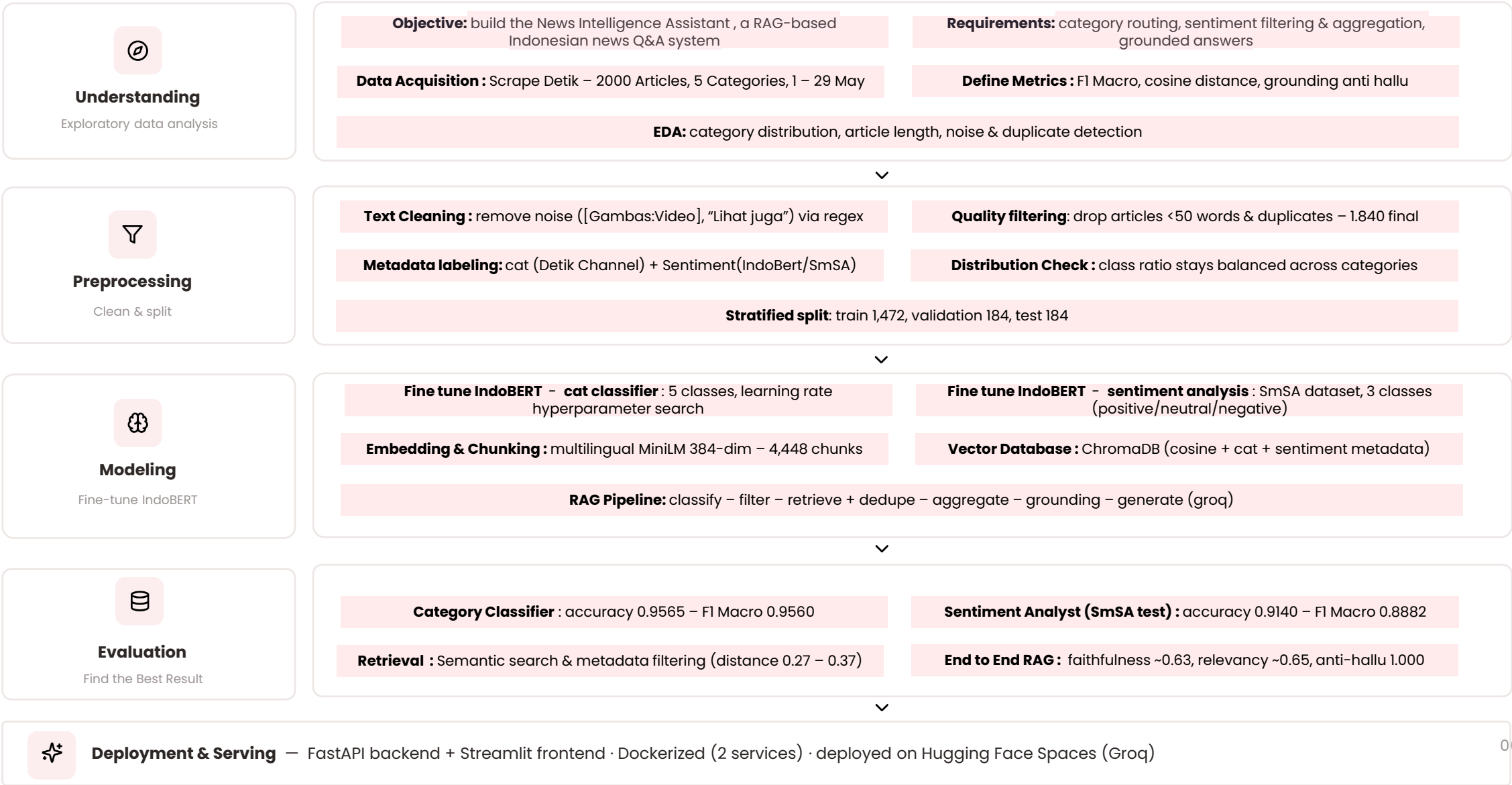


Krisna Fery Rahmantya

GROUP C · NLP

From raw articles to *grounded answers*

An end-to-end pipeline — each stage feeds the next, from news-corpus construction through to a deployed, grounded question-answering interface.



The stack, *at a glance*



Docker
Containerization



ChromaDB
Vector database



Hugging Face

Hugging Face
Models · Spaces



FastAPI
Backend API



Streamlit
Frontend UI



PyTorch
PyTorch
Model training



Groq
LLM (default)



Gemini
LLM (planned)

PART 01

Understanding (*exploratory data analysis*)

Scraping, exploring, and cleaning 2,000 Indonesian news articles into a balanced, model-ready corpus.

01

Scraping the *Detik* news corpus

A two-stage crawler with **requests** + **BeautifulSoup** builds the knowledge base: collect article URLs from paginated indexes, then parse each article — with rate limiting and URL validation throughout.

2,000

ARTICLES COLLECTED

The raw corpus assembled from five Detik channels.

5 × 400

BALANCED BY CHANNEL

news · finance · sport · inet · health

1–29 May

COVERAGE WINDOW · 2026

Defines the temporal boundary of the RAG knowledge base.

articles.csv

OUTPUT ARTEFACT

Title, body, category, URL, and date metadata.

CRAWL PROCEDURE

1 Collect URLs

Walk paginated indexes (`?page=N`) to gather valid article links.

2 Parse articles

Extract title, body, and metadata per page, with rate limiting and URL filtering.

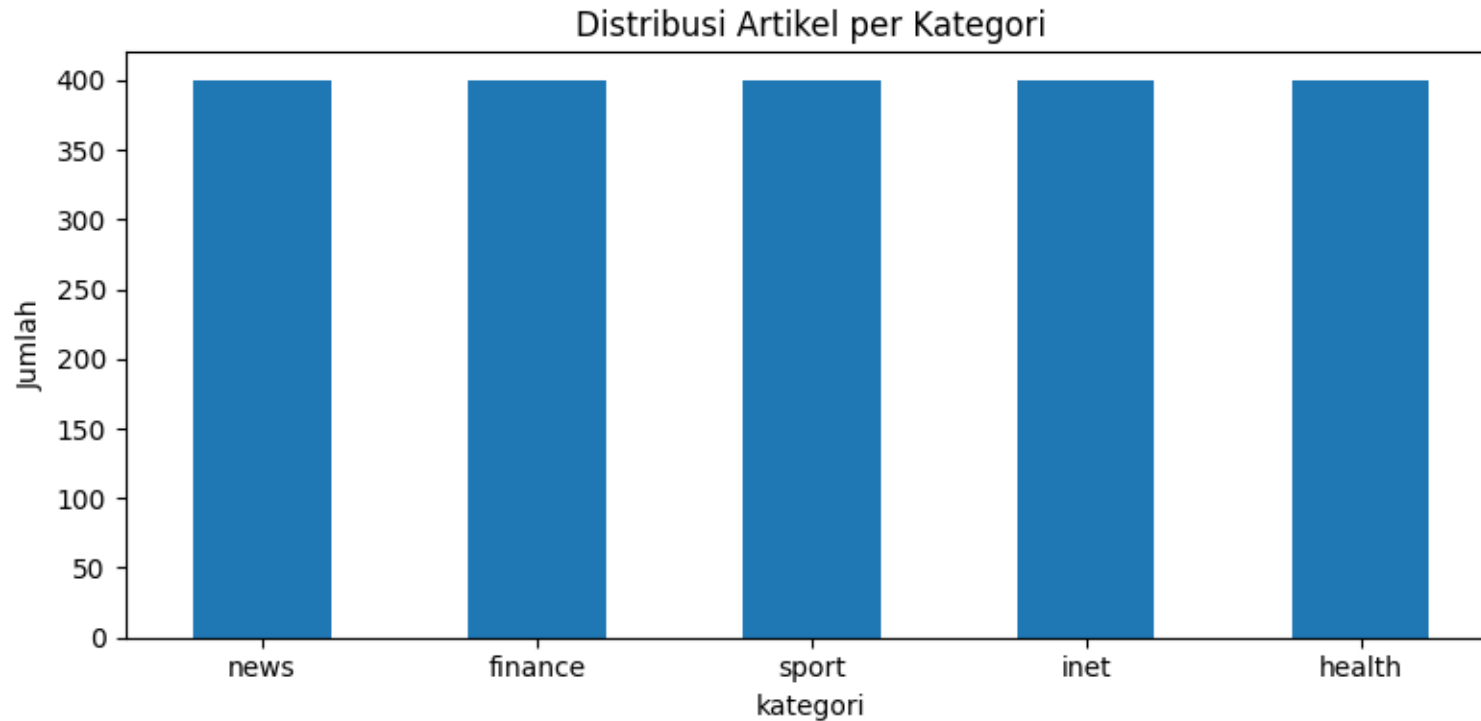
🌐 requests

<> BeautifulSoup

📖 Source: Detik.com — channels news, finance, sport, inet, health · 1–29 May 2026

A perfectly balanced corpus

Each of the five channels contributes exactly **400 articles** — a min/max ratio of 1.00. No dominant class, which is ideal for training an unbiased category classifier.



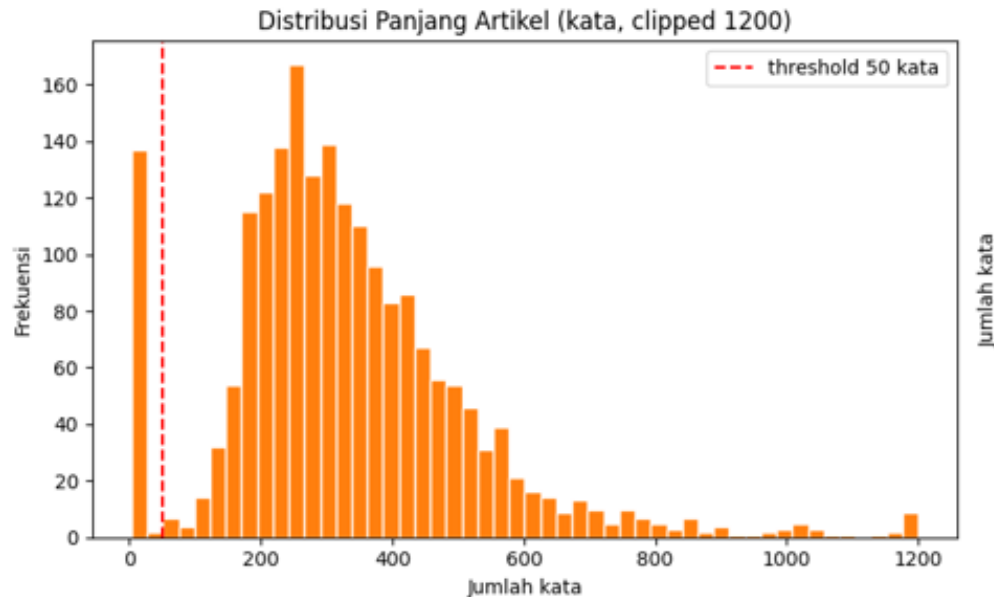
1.00

BALANCE RATIO (MIN / MAX)

Perfectly uniform across all five categories
— the classifier learns each class on equal footing, with no resampling required.

A right-skewed length distribution

Article word counts cluster around a few hundred words with a long upper tail. A natural gap in the distribution sets the **50-word filtering threshold** that removes thin, low-content pages.



~333

MEAN WORDS · MEDIAN 304

139

ARTICLES BELOW 50 WORDS

Removed — mostly photo galleries & placeholders.

sport ↔ inet

SHORTEST ↔ LONGEST CHANNELS

What needs cleaning — and how



Residual noise

Video markers [Gambas:Video] appear in 20.7% of articles (414); "Lihat / Simak juga Video" in ~6% — embedded mid-text, so they must be stripped.



Duplicates

No duplicate article IDs or URLs, but 47 bodies (62 rows) and 2 titles repeat — all within the same category. Removed keep-first.



Dates & limits

4 empty dates (0.2%) are left as-is. The 1–29 May 2026 range becomes the RAG knowledge boundary.

PREPROCESSING PLAN

- 1 Remove non-content noise (markers, "Baca juga")
- 2 Normalize excess whitespace
- 3 Recompute word counts after cleaning
- 4 Drop articles under 50 words (~139)
- 5 Remove duplicate bodies (~47) & titles (~2), keep-first
- 6 Leave empty dates untouched



No aggressive stemming or stopword removal — **IndoBERT is contextual & case-aware**, and RAG needs natural, intact text.

PART 02

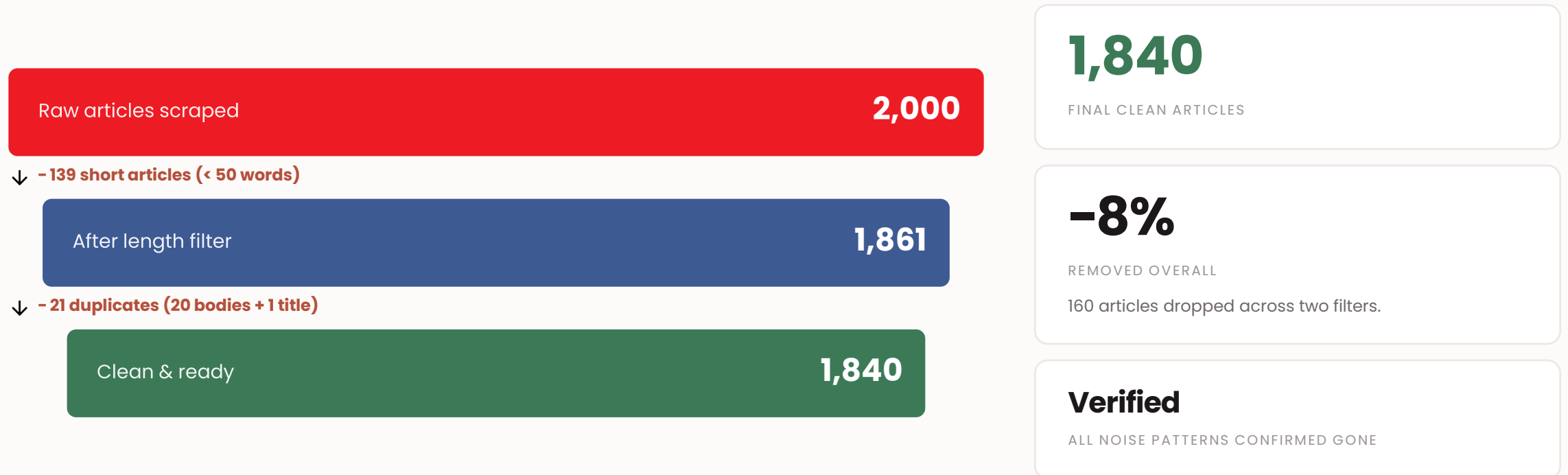
Preprocessing (*cleaning the noise*)

Refining 2,000 Indonesian news articles into a clean, balanced, and model-ready corpus

02

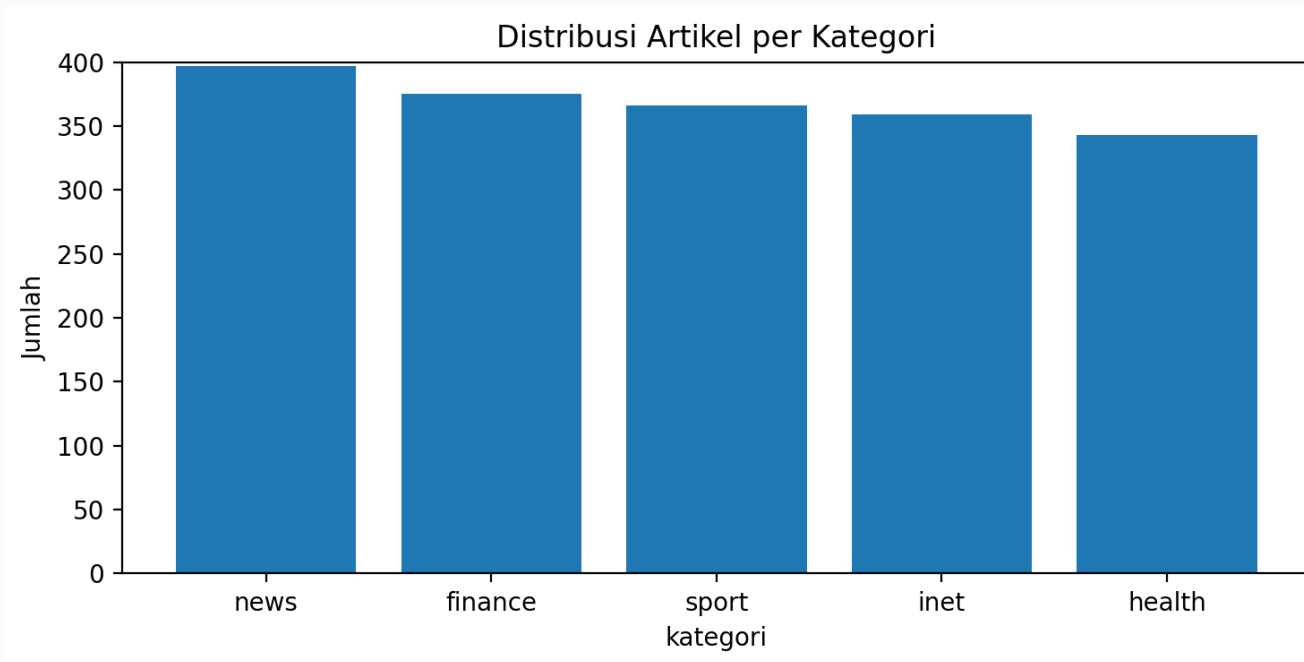
From 2,000 raw to *1,840 clean* articles

Regex strips video markers, "Baca juga", and related-link noise; thin and duplicate articles are dropped keep-first. An 8% reduction yields a verified, ready-to-use corpus.



Still balanced after cleaning

Cleaning shifts the per-category counts only slightly — every channel stays within **343–397 articles**, with no dominant class. news ends largest, health smallest.



① Per-category counts representative within the reported 343–397 range · total 1,840

265–400

MEAN WORDS · SPORT → INET

› 50 words

MINIMUM LENGTH, EVERY CATEGORY

Stratified 80 / 10 / 10 split

The corpus is partitioned by category with a fixed random seed, so each class keeps a consistent proportion across train, validation, and test — fully reproducible.



HOW IT'S SPLIT

Stratified by category with a fixed `random_state` for reproducibility, preserving class proportions in every partition.

T

No aggressive stemming, stopword removal, or lowercasing — IndoBERT is contextual & case-aware.

OUTPUT ARTEFACTS

📄 `articles_clean.csv` · 1,840

📁 `data/splits/train.csv · val.csv · test.csv`

→ Ready to train the category classifier — the routing component of the RAG system.

PART 03

Modeling (***fine tuned –language models***)

A shared IndoBERT foundation powers category routing and sentiment metadata — the intelligence layer beneath retrieval.

03

Role & Data

Supplementary metadata for opinion filtering and aggregation in RAG systems



Role 1 – Filter

Role 2 – Opinion aggregation Filtering articles based on sentiment, e.g. answering a query for "positive news" on a topic.



Role 2 – Opinion aggregation

Summarizes the sentiment trend on a topic or category (e.g., the majority of news is negative).

DATA & LABEL

11.000 / 1.260 / 500

training/validation/test samples (IndoNLU SmSA)

3 classes: positive, neutral, negative

Source: IndoNLU SmSA review – used because there is no news dataset labeled with sentiment (different from category).max_length 128 tokens (short review text).

One architecture, *two models*

Both the sentiment and category classifiers fine-tune the same backbone — **IndoBERT (indobert-base-p1)** — giving the system a single, consistent, and maintainable foundation.



Why IndoBERT

Pre-trained on the large Indo4B Indonesian corpus, so it grasps context, idiom, and morphology far better than generic multilingual models.



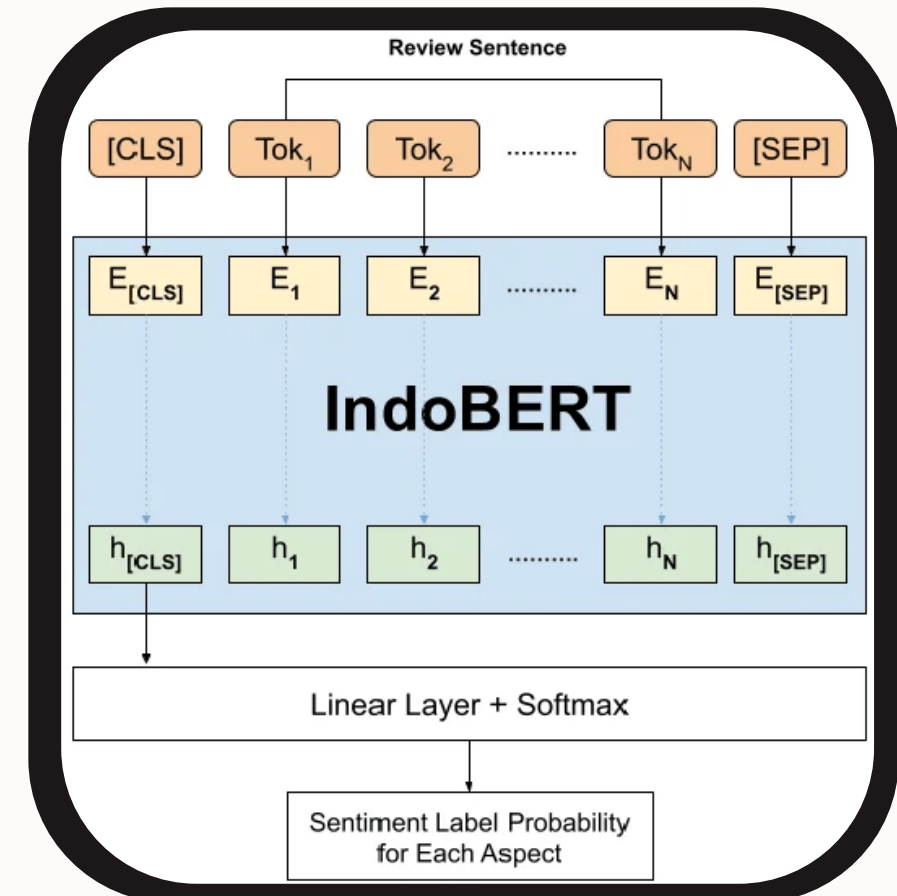
Why the BERT architecture

A bidirectional Transformer yields contextual embeddings — word meaning depends on the sentence. Fine-tuning delivers strong transfer learning even with limited labels.



Classification head

The [cls] token representation feeds a lightweight classifier for both tasks.



Sentiment as *complementary metadata*

Sentiment enriches retrieval in two ways — filtering and opinion aggregation. With no labelled news-sentiment data available, the model is trained on the **IndoNLU SmsA** review corpus.

01 Filtering

Surface articles by sentiment — e.g. answer a request for "positive news" on a topic.

02 Opinion aggregation

Summarise the sentiment tendency of a topic or category — e.g. mostly negative coverage.

WHY SMSA, NOT SCRAPED DATA

- × Scraped articles carry no sentiment labels at all
- × Manually labelling 1,840 articles is infeasible & inconsistent
- × No labelled Indonesian news-sentiment dataset at scale

INDONLU SMSA DATASET

11,000

train

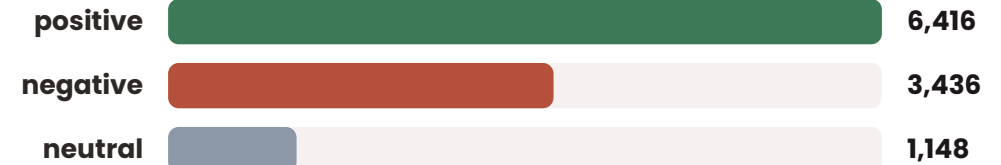
1,260

validation

500

test

CLASS DISTRIBUTION (IMBALANCED)



3 classes · positive · neutral · negative · max_length **128** · eval with **F1-macro**

Disciplined hyperparameter search

Only the learning rate is tuned — across the three values recommended by Devlin et al. (2019) — while epoch, batch size, and sequence length stay fixed at paper-recommended defaults.

TUNED

Ir · 2e-5

Ir · 3e-5

Ir · 5e-5

FIXED

epoch · 3

batch · 16

max_length · 128

seed · 42

WHY ONLY THE LEARNING RATE?

- ✓ LR most determines fine-tuning quality & stability
- ✓ A full grid (18 runs) is costly on a single GPU
- ✓ Batch & epoch locked to BERT-paper recommendations

BEST CONFIGURATION

Ir 2e-5

epoch 3 · batch 16 · max_length 128 · seed 42

0.9413 **0.9150**

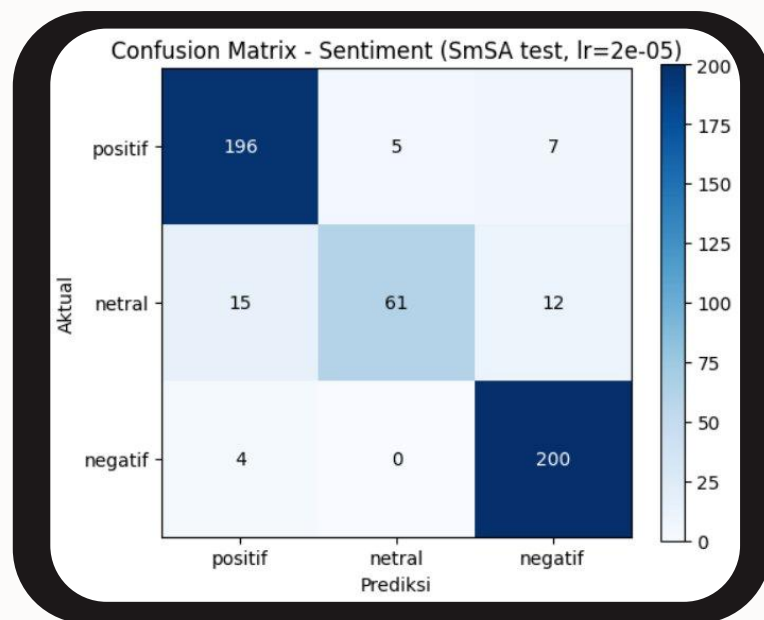
Validation accuracy Validation F1-macro

Selected via load_best_model_at_end

🔗 Hyperparameter values follow Devlin et al. (2019) · arxiv.org/abs/1810.04805

Strong overall, weakest on *neutral*

On the 500-sample SmSA test set the model reaches solid accuracy. The neutral class lags on recall — a minority class (88 samples) and inherently ambiguous, so some cases drift to positive or negative.



0.9140

TEST ACCURACY

0.8882

TEST F1-MACRO

Macro averaging reflects the class imbalance.

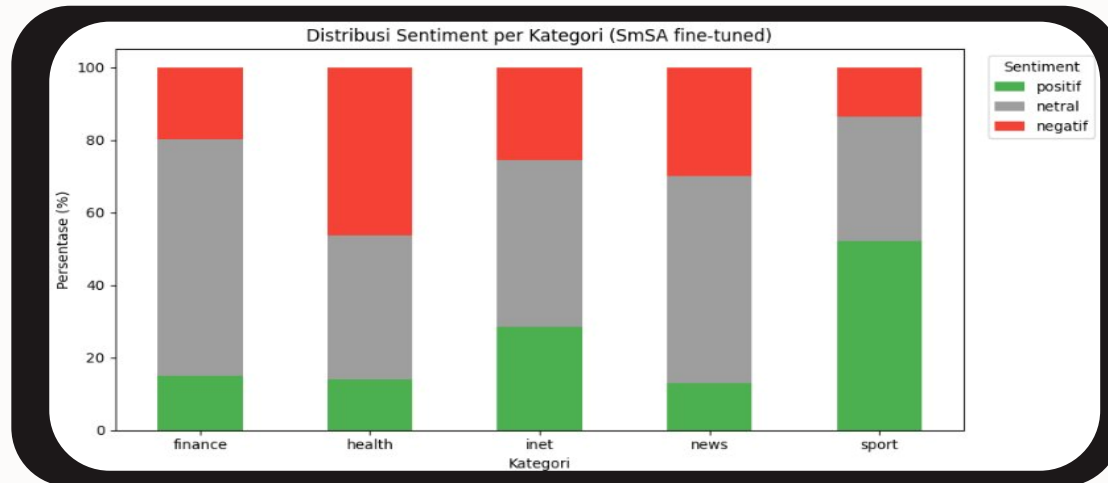
CLASS	PRECISION	RECALL	F1
positive	0.91	0.94	0.93
neutral	0.92	0.69	0.79
negative	0.91	0.98	0.95



Neutral recall of **0.69** is the headline weakness — expected given only 88 neutral samples and the genuinely fuzzy boundary between neutral and polar sentiment.

Labelling all 1,840 articles

The model labels every clean article (up to 512 tokens). Results follow intuition — sport skews most positive, health most negative, finance and news mostly neutral.



898

neutral

494

negative

448

positive

OUTPUT

 articles_with_sentiment.csv

Sentiment label + score attached to every article for downstream filtering.

SAMPLE LABELS

"Megawati Hangestri Pertiwi resmi gabung Hyundai Hillstate untuk musim 2026/2027."

Achievement story → correct

sport · positive

score 1.00

"IHSG Dibuka Dua Arah, Balik Menguat ke 6.148"

Factual report → correct

finance · neutral

score 1.00

"MotoGP: Diggia Jatuh Dihantam Ban, Juara..."

Rider actually won, but "jatuh / dihantam" reads negative → domain shift

sport · negative

score 0.58

The routing brain of retrieval

The category model both tags articles at index time and routes user questions at run time. Labels come free from the Detik channels — no external dataset or manual annotation required.

01 Category metadata

Tags each article during knowledge-base preparation, so retrieval can be filtered per category.

02 Query routing — run time

Classifies the user's question and steers search to the relevant category. Its accuracy directly drives the relevance of retrieved articles.

TRAINING DATA

1,472

train

184

validation

184

test

finance

health

inet

news

sport



Labels are inherited from Detik channels — and **max_length 256** gives richer context for the harder routing task.

Training: Hyperparameters and Selected Parameters

Search for the best learning rate and configuration (same procedure as sentiment model)

01 Tuned vs fixed

Hypertune: learning rate -> 2e-5, 3e-5, 5e-5.

Fixed: epoch 3, batch 16, max_length 256, seed 42.

02 Why only learning rate?

- LR is the most important factor in determining the results and stability of fine-tuning.
- A full grid (LR x batch x epoch = 18 runs) is expensive on a single GPU.
- Batches and epochs are locked to the recommended values in the BERT paper.
- The three LR values are exactly as suggested by Devlin et al. (2019), not random guesses.

BEST CONFIGURATION

lr 2e-5

best learning rate (of the 3 tested, F1-macro val 0.935 - 0.941)

Epoch 3 · Batch 16 · Max_length 256 · Seed 42

0,9402

0,9407

Validation accuracy

Validation F1-macro

Selected via load_best_model_at_end

Near-perfect, almost diagonal

Same procedure as the sentiment model — learning rate tuned, everything else fixed. On the 184-sample test set the classifier is highly accurate; news scores highest, inet slightly lower where it overlaps business and science.

0.9565

TEST ACCURACY

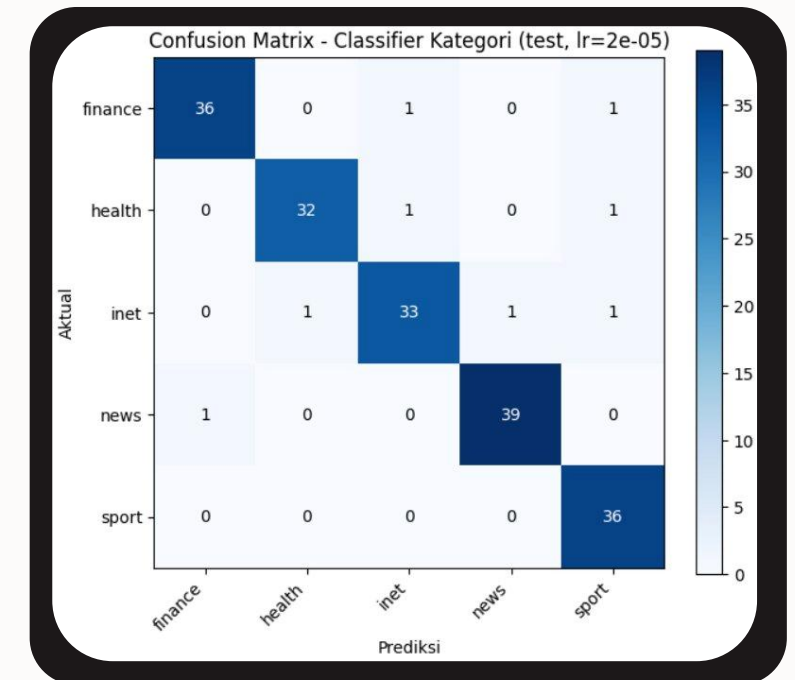
0.9560

TEST F1-MACRO

Best **lr 2e-5** · epoch 3 · batch 16 · max_length 256 · val F1-macro 0.9407

|| Per-class F1 on the held-out test set · max value 1.00

Class	P	R	F1
finance	0,97	0,95	0,96
health	0,97	0,94	0,96
inet	0,94	0,92	0,93
news	0,97	0,97	0,97
sport	0,92	1,00	0,96



Confidence becomes a *routing signal*

At run time each question is classified into a category with a confidence score. Specific questions route firmly; broad questions stay cautious so retrieval remains relevant.

USER QUESTION	CATEGORY	CONFIDENCE
"Kabar ekonomi & pergerakan saham terbaru?"	finance	0.94
"Spesifikasi & harga smartphone flagship terbaru"	inet	0.94
"Tips menjaga kesehatan jantung & pola makan"	health	0.89
"Hasil pertandingan & klasemen sepak bola pekan ini"	sport	0.81
"Bagaimana antusiasme umat muslim Idul Adha?"	news	0.65

HIGH CONFIDENCE

Specific queries → category filter applied firmly.

MODERATE CONFIDENCE

Broad queries (Idul Adha → news 0.65) → the system applies filters more cautiously to keep retrieval relevant.

PART 04

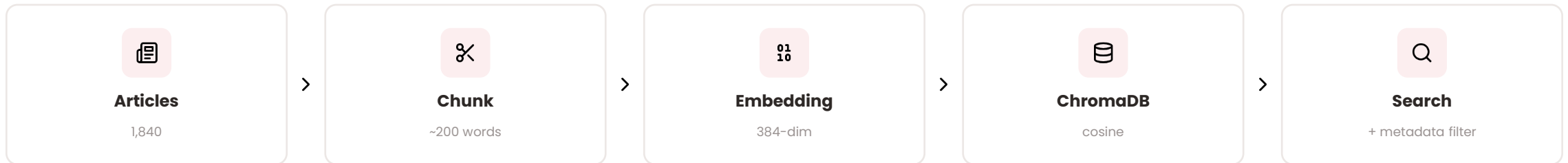
Vector DataBase

A semantic vector database that serves as the foundation for transforming data into a grounded, source-cited assistant. It emphasizes how live, adaptive behaviour enhances the system's trustworthiness and reliability.

04

Searching by *meaning*, not keywords

Each article is chunked, embedded into a 384-dimension vector, and stored in ChromaDB with its text and metadata — so "saham naik" and "IHSG menguat" land close together even with no shared words.



Semantic search

Compares vectors, not keywords — similar meaning is retrieved even when the wording differs.



Three parts per entry

Vector for search + the original chunk text + metadata (category, sentiment, title, URL).



Why ChromaDB

Lightweight, server-less, persistent to disk; cosine ranking via an HNSW graph index (fast approximate nearest-neighbour) with metadata filtering.

Why we chose *ChromaDB*



VECTOR STORE & COSINE

The store holds all 4,448 embeddings and ranks them by **cosine similarity** — the angle between vectors, not their length — so meaning matches even with no shared words; an HNSW index keeps nearest-neighbour search fast.

TRADE-OFFS

- ✓ Single-node, in-process — no horizontal scaling
- ✓ Not built for millions of vectors or heavy concurrency
- ✓ Younger / less battle-tested than enterprise stores
- ✓ At our scale (4.4k chunks), that ceiling never bites

WHY IT FITS THIS PROJECT

- ✓ Embedded & serverless — no separate DB process
- ✓ Bundles straight into the Docker image / HF Space
- ✓ Native metadata filters power category + sentiment routing
- ✓ Python-native, free, effortless at 4,448 vectors

WHY NOT THE OTHERS

- ✓ Pinecone — managed, paid, external dependency
- ✓ FAISS — a library, not a DB; no built-in metadata
- ✓ Qdrant / Weaviate — a separate server to run
- ✓ Milvus — distributed infra; overkill here

Splitting articles into *retrievable units*

Whole-article vectors blur meaning and blow past token limits. Articles are split into sentence-aware chunks of ~200 words with a one-sentence overlap, so context never breaks mid-thought.



Why split?

Precision — one article as one vector averages away meaning. Plus embedding/LLM token limits, and context efficiency: only relevant chunks reach the LLM.



How?

Target ~200 words (ideal 150–250 for news), a one-sentence overlap between chunks, and sentence-aware boundaries that never cut mid-sentence.

4,448

CHUNKS FROM 1,840 ARTICLES

2.42

avg chunks / article

~158

avg words / chunk

Short articles stay one chunk; long ones split into several. Most chunks come from news & inet, fewest from sport.

Text becomes a *384-dim vector*

Every chunk is encoded into a 384-number vector that captures its meaning; semantically similar text lands nearby. The same model, sentence-transformers MiniLM-L12-v2, encodes both the index and the queries.



Meaning as geometry

Similar text → nearby vectors, which is what makes the search semantic rather than lexical.



Key practices

One model for index & query: sentence-transformers paraphrase-multilingual-MiniLM-L12-v2 (384-dim, multilingual), with normalised vectors for cosine similarity.

STORAGE

4,448 × 384

vector matrix · chunks × dimensions

- ChromaDB PersistentClient · cosine (hnsw:space)
- Batched inserts of 500 → all 4,448 chunks
- Persisted to chroma_db/ — survives restarts

Semantic search, made *context-aware*

Category and sentiment become where filters on top of pure semantic search — the integration point where the two classifiers meet retrieval.

(A) NO FILTER · PURE SEMANTIC

"kabar ekonomi dan pergerakan saham IHSG"

top-3 all finance

distance 0.33–0.37

(B) CATEGORY = INET

"gadget dan teknologi terbaru"

3 chunks · all inet

top: "Erafone Hadir Lebih Dekat"

(C) CATEGORY SPORT + POSITIVE

"pertandingan olahraga Indonesia"

3 sport · positive

distance 0.27–0.28



Classifier category & sentiment become metadata filters → retrieval is context-aware. **Note:** one long article can yield several chunks, so the pipeline de-duplicates per article downstream.

PART 05

Retrieval-Augmented Generation

A method that transforms a semantic vector store into a grounded, source-cited assistant. It highlights how live, adaptive behaviour contributes to the system's trustworthiness and reliability.

05

Answers anchored to *real data*

RAG pairs the knowledge base with the LLM: retrieve relevant chunks, augment the prompt with them, and generate an answer drawn only from that context — with citations.



RETRIEVAL

The question is embedded with the same model; nearest chunks are found in ChromaDB by cosine and de-duplicated per article.



AUGMENTED

The relevant chunks are injected into the LLM prompt as context, giving the model real material to answer from.



GENERATION

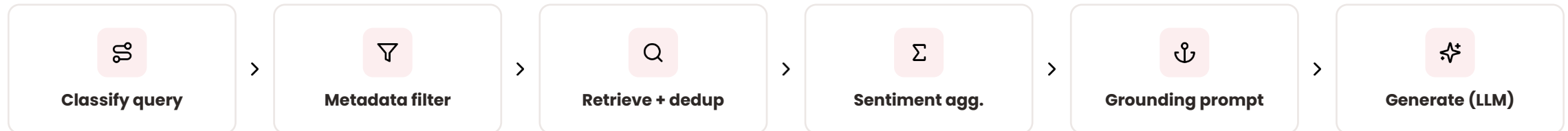
The LLM writes a natural-language answer only from the supplied context, with numbered source citations.



Grounding — answer **only** from context — makes sources traceable and keeps the system honest when information is missing. This anti-hallucination behaviour is RAG's edge over a plain LLM.

Orchestrating the system at *run time*

A single function chains classification, filtered retrieval, sentiment aggregation, grounding, and generation — with sensible fallbacks at every step.



 **Routing & fallback**
Filter to the predicted category only when confidence ≥ 0.5 ; otherwise search stays broad. Too few results → retry without the filter.

 **Optional sentiment**
A sentiment filter is added only when the user explicitly asks for it (e.g. "positive" news).

 **Grounding & efficiency**
Grounding prompt, per-article dedup, temperature 0.3. Groq Llama 3.3 70B by default; Gemini 2.5 Flash optional.

The LLM layer — *Groq by default*



PROVIDERS & MODELS

Generation runs on Groq by default — model **llama-3.3-70b-versatile**. Google Gemini (gemini-2.5-flash) is selectable as an alternative. The provider is chosen per request; the API key is read from the environment, never the request body.

GENERATION SETTINGS

- ✓ temperature = 0.3 (focused, low randomness)
- ✓ System prompt: answer from the context only
- ✓ Must cite source numbers, e.g. [1], [2]
- ✓ Empty context → honest "tidak ditemukan"

RELIABILITY

- ✓ Retry with backoff (1, 2, 4 s) on 503 / 429
- ✓ Up to 4 attempts before failing
- ✓ Daily-quota errors surfaced clearly
- ✓ Models loaded once at startup; key from ENV

WHY GROQ BY DEFAULT

- ✓ Fast inference, low latency
- ✓ Generous free daily token budget
- ✓ The only provider evaluated end-to-end
- ✓ Gemini returns after RAGAS evaluation

Routing and filtering, *in action*

EXAMPLE 1 · CATEGORY SUMMARY

"Berita teknologi atau gadget terbaru apa saja?"

category **inet · 0.85**

filter **inet**

2 positive · 3 neutral

- 1 Erafone launches its "Lebih Dekat" campaign for easier gadget shopping.
- 2 Telkomsel outlines an AI-based digital-company strategy (Autonomous Network, Veronika, Hyper 5G).
- 3 The "Drachin Pendek" micro-drama trend is tipped as a digital-entertainment phenomenon.

EXAMPLE 2 · SENTIMENT FILTER

"Ada kabar olahraga Indonesia yang membanggakan?"

category **sport · 0.97**

filter **sport + positive**

5 positive sources

- 3 Indonesia hosts the ADCC, the world's most prestigious grappling competition.
- 4 Rafting athletes win gold, silver & bronze at the IRF World Rafting Championship 2025.
- 5 Fajar / Shihibul Fikri take the China Open Super 1000 title.

When the answer isn't there, it *says so*

EXAMPLE 3 · NOT FOUND

"Berapa harga tiket konser Coldplay di Jakarta?"

"Saya tidak menemukan informasi mengenai harga tiket konser Coldplay di Jakarta dalam artikel yang tersedia."

A few loosely "tiket"-related chunks were retrieved, but the LLM declined to invent an answer.



WHY IT MATTERS

Grounding works: the system **admits its limits** instead of fabricating a convincing but false answer.

This is trustworthy RAG behaviour — the core advantage over an ungrounded LLM. The pipeline is complete and proven, ready to ship as an app.

Grounded by design — and *measured for it*

Automatic, reference-free evaluation over 100 questions — 55 across the 5 categories (11 each), 33 sentiment-filtered (11 per sentiment: positive / neutral / negative), and 12 out-of-corpus. An LLM judges how grounded and relevant each answer is, plus whether refusals are correct; the next slide breaks these scores down.



1.000

ANTI-HALLUCINATION

Every "not found" refusal was correct — the system never fabricated for out-of-coverage questions.



~0.63

FAITHFULNESS

Answer claims are generally supported by the retrieved context (overall, across 91 answered questions).



~0.65

ANSWER RELEVANCY

Answers address the question directly. Strongest on health, weakest on inet category.



Judged with an **LLM-as-a-judge (Groq)**, similar to **RAGAS** — reference-free, incremental, and quota-aware.

By category and *sentiment*

BY CATEGORY

CATEGORY	Faith	Rel	Anti	n
health	0.758	0.753	1.000	20
sport	0.667	0.722	1.000	21
finance	0.595	0.641	—	22
news	0.588	0.594	1.000	17
inet	0.525	0.525	1.000	20

n = all 100 answers grouped by the classifier's predicted category (sums to 100), not the 11-per-category design.

BY SENTIMENT

SENTIMENT	Faith	Rel	Anti	n
negative	0.682	0.736	—	11
neutral	0.460	0.490	1.000	11
positive	0.460	0.410	1.000	11
no-filter*	0.677	0.703	1.000	67

*no-filter = overall baseline, shown for comparison

Strengths: health and negative-tone news score highest — fact-dense, explicit text grounds easily; anti-hallucination is perfect (1.000) wherever the system refused.

Weaknesses: inet and positive/neutral score lowest — thinner, more abstract content is harder to ground, and adding a filter narrows the context and nudges scores down.

PART 06

Deployment, Demo & Conclusion

App demo, conclusions, and future works

06

A clean split — *backend & frontend*

THE SPLIT

The RAG engine and the UI are separate services. The **backend (FastAPI)** does all the work; the frontend (Streamlit) only calls it over HTTP — so the UI stays light (no model, no GPU).

BACKEND (FASTAPI)

- ✓ Loads models once at startup (lifespan), kept in memory
- ✓ Endpoints: GET /health and POST /ask
- ✓ Pipeline: classify → filter → retrieve → generate
- ✓ API key read from ENV, never from the request body

FRONTEND (STREAMLIT)

- ✓ Pure UI — no torch, no chromadb, no GPU
- ✓ Calls the backend via BACKEND_URL over HTTP
- ✓ Sidebar filters: category, sentiment, top-k
- ✓ Shows answer, sources, and a sentiment bar

WHY THE SPLIT

- ✓ Restart the UI without reloading heavy models
- ✓ Backend reusable by any client, not just Streamlit
- ✓ Lighter, faster frontend deploys
- ✓ Same engine runs locally and on Hugging Face

Two services, *one command*

docker compose builds and runs the backend and frontend together. Each service is its own container; ChromaDB and the models are mounted as volumes so the image stays reproducible.



Two services

backend (FastAPI, 8000) and frontend (Streamlit, 8501) build from their own Dockerfiles and start with one docker compose up.



Volumes & networking

chroma_db mounted read-write (needs lock/WAL), models read-only. The frontend reaches the backend by service name (`http://backend:8000`), not localhost.

COMPOSE

2 containers

one network · compose-orchestrated

- backend: FastAPI + Uvicorn · port 8000
- frontend: Streamlit · port 8501
- volumes: chroma_db (rw) · models (ro)

One container, *public on HF*

The same system ships to Hugging Face Spaces as a single Docker container: FastAPI runs internally while Streamlit is exposed publicly. The RAG pipeline, models, and ChromaDB are identical to local.

GITHUB REPOSITORY (LOCAL)

<https://github.com/KamigonNoMercy/indonesian-news-intelligence-apps>

ASPECT	LOCAL	HF SPACES
Orchestration	2 containers	1 container
Public port	8000 + 8501	7860 only
LLM provider	Groq or Gemini	Groq only
torch build	GPU or CPU	CPU-only
Model + vector DB	host volumes	baked (git-lfs)

WHY GROQ-ONLY

Only Groq is evaluated end-to-end, so the public demo matches measured behavior.

LIVE DEMO (HF SPACES)

<https://github.com/KamigonNoMercy/indonesian-news-intelligence-apps>

The full app, *live on Docker*

The whole pipeline runs locally via docker-compose — two containers (FastAPI + Streamlit). A walkthrough of one real query, end to end.



User input

A question box plus an optional sentiment filter (positive / neutral / negative). One click runs `rag_answer()` behind the scenes.



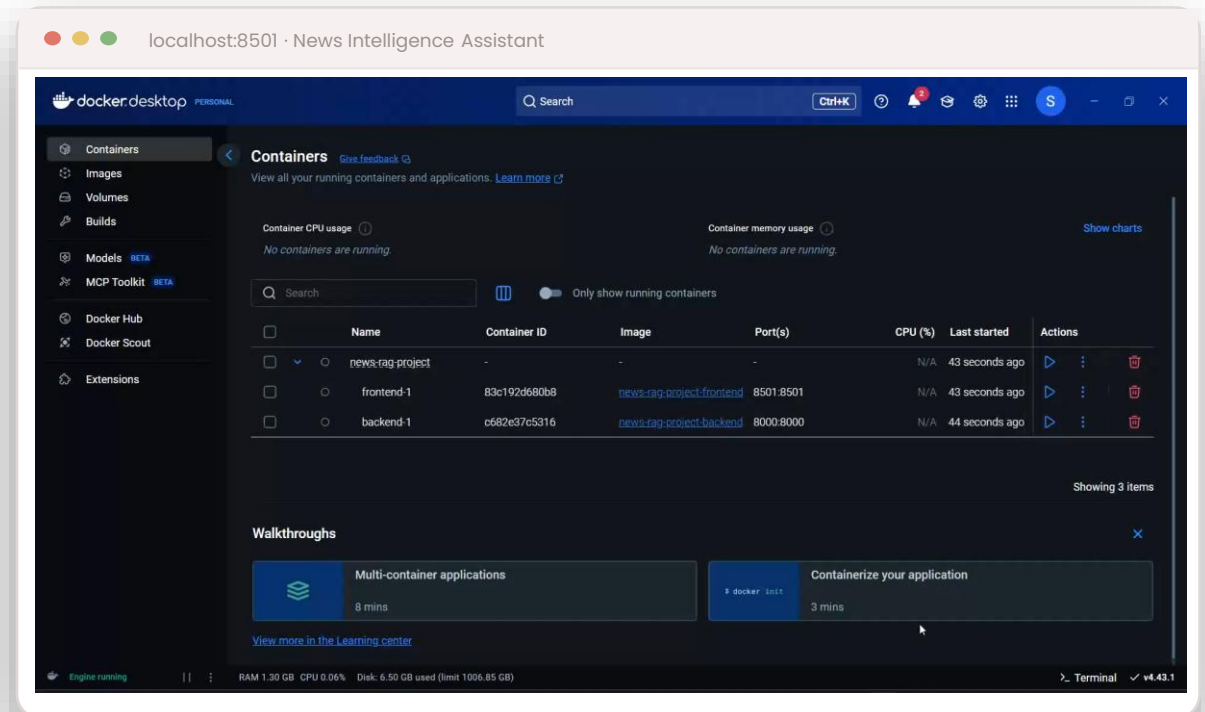
Result view

Shows the detected category, the filters applied, a sentiment summary, the cited answer, and a source list with original links.



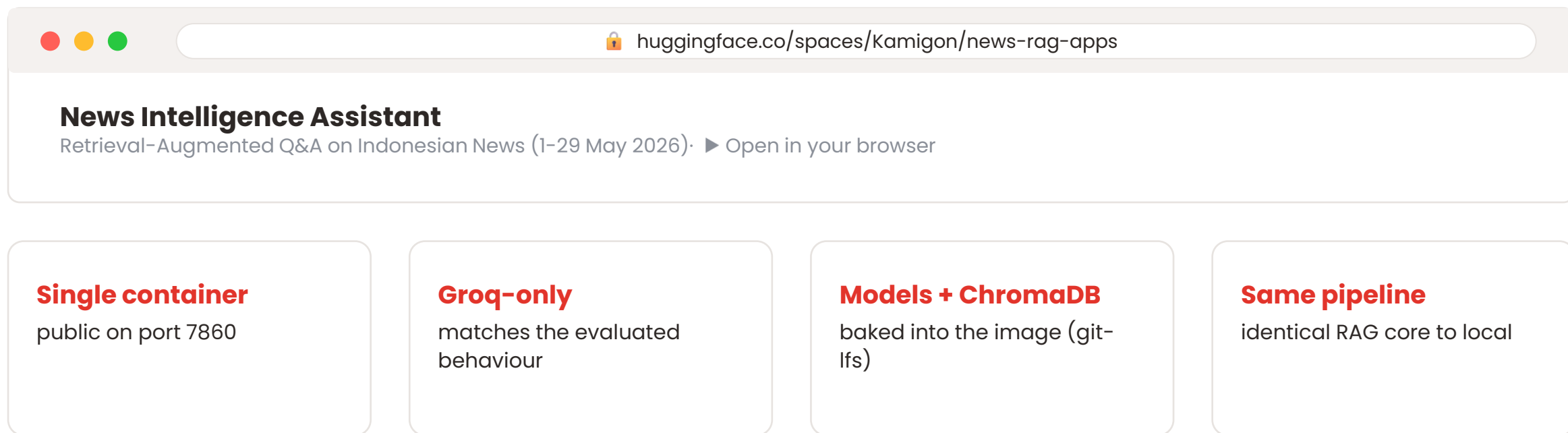
"Not found" handling

When information is absent, only the honest answer is shown — category, sentiment, and sources are hidden to avoid confusion.



Live & public — *on HF Spaces*

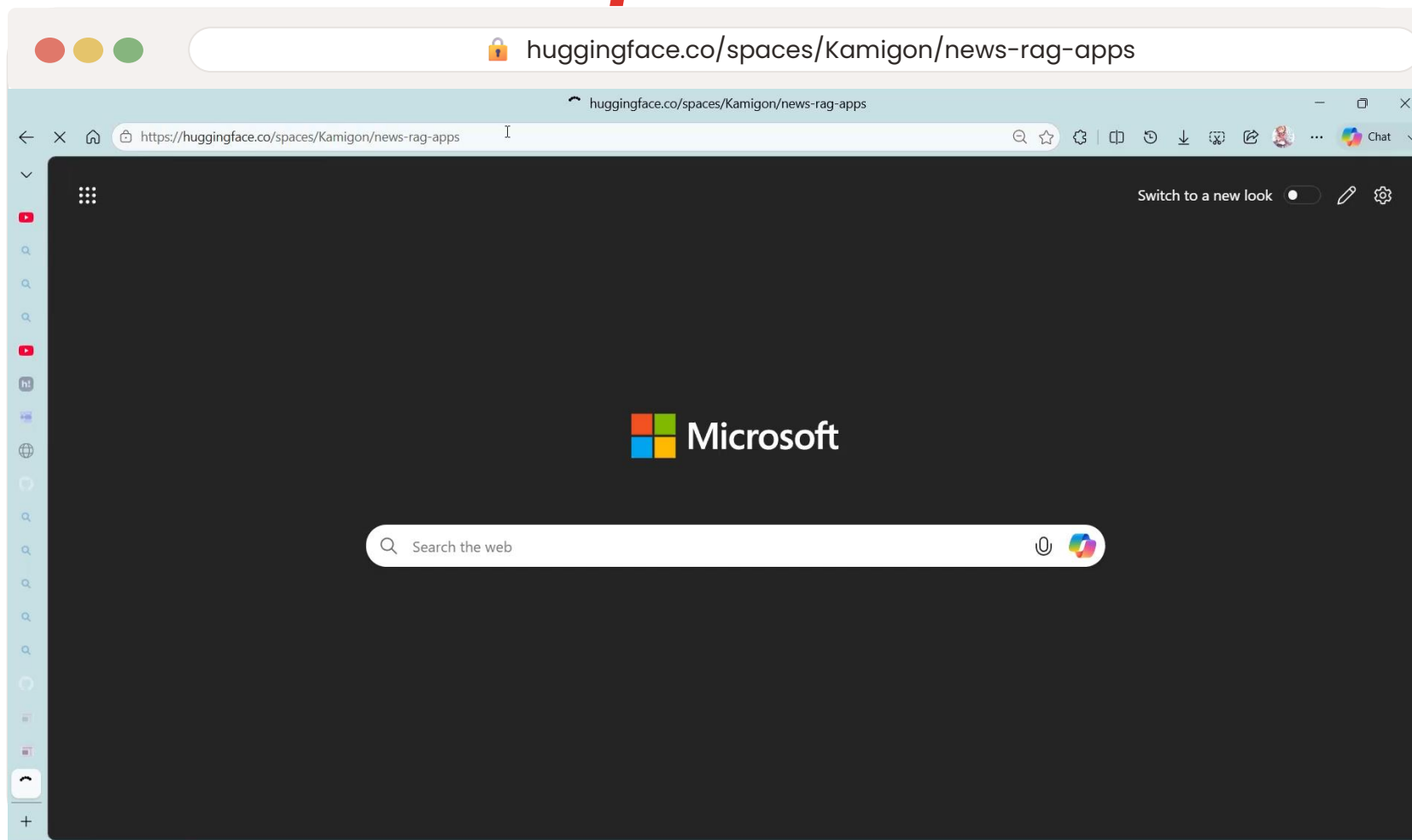
The same app, deployed as a single-container Space — anyone can open it in the browser, no setup. Identical RAG pipeline, models, and ChromaDB to the local version.



The screenshot shows a web browser window with the address bar displaying `huggingface.co/spaces/Kamigon/news-rag-apps`. The page title is **News Intelligence Assistant**, with a subtitle *Retrieval-Augmented Q&A on Indonesian News (1-29 May 2026)* and a link [► Open in your browser](#). Below the header, there are four feature boxes:

- Single container**
public on port 7860
- Groq-only**
matches the evaluated behaviour
- Models + ChromaDB**
baked into the image (git-lfs)
- Same pipeline**
identical RAG core to local

Live & public — *on HF Spaces*



A complete, grounded system — and where it goes next

WHAT WE DELIVERED

- ✓ End-to-end RAG news Q&A: scrape → preprocess → fine-tune IndoBERT → vector DB → pipeline → app.
- ✓ Three integrated components: category classifier (test acc 0.96) for routing, sentiment for filter & aggregation, retrieval + generation at the core.
- ✓ Answers grounded in real data, source-cited, with anti-hallucination 1.000.
- ✓ Clean BE/FE split — local Docker + live on HF Spaces.

FUTURE WORK

- **Re-enable Gemini:** evaluate it with the RAGAS pipeline, then restore the provider switch on HF.
- **Weak segments:** lift the inet category and positive/neutral sentiment — clean/expand those slices, revisit chunking.
- **Data labeling & coverage:** recurring scrape beyond one month; more sources, wider date range, also label the sentiment instead using SMSA
- **Reranking & hardening:** cross-encoder reranking (BM25 + semantic); tighten CORS, logging, rate limiting, caching.

NEWS INTELLIGENCE ASSISTANT

Thank *you*

Questions & discussion welcome — let's talk about
grounded, source-cited news intelligence.

Group C · Batch 10 NLP | June 2026

